

SHETH L.U.J & SIR M.V COLLEGE OF SCIENCE

SUBJECT : CYBER SECURITY

AIM : Set up and configure an intrusion detection system (IDS) to monitor network traffic and detect potential security breaches or malicious activities.

Step 1: Update Ubuntu

Open the Terminal and run:

```
sudo apt update
```

```
sudo apt upgrade -y
```

Step 2: Install Snort

Install Snort using:

```
sudo apt install snort -y
```

During installation, Snort may ask for the local network range. Example:

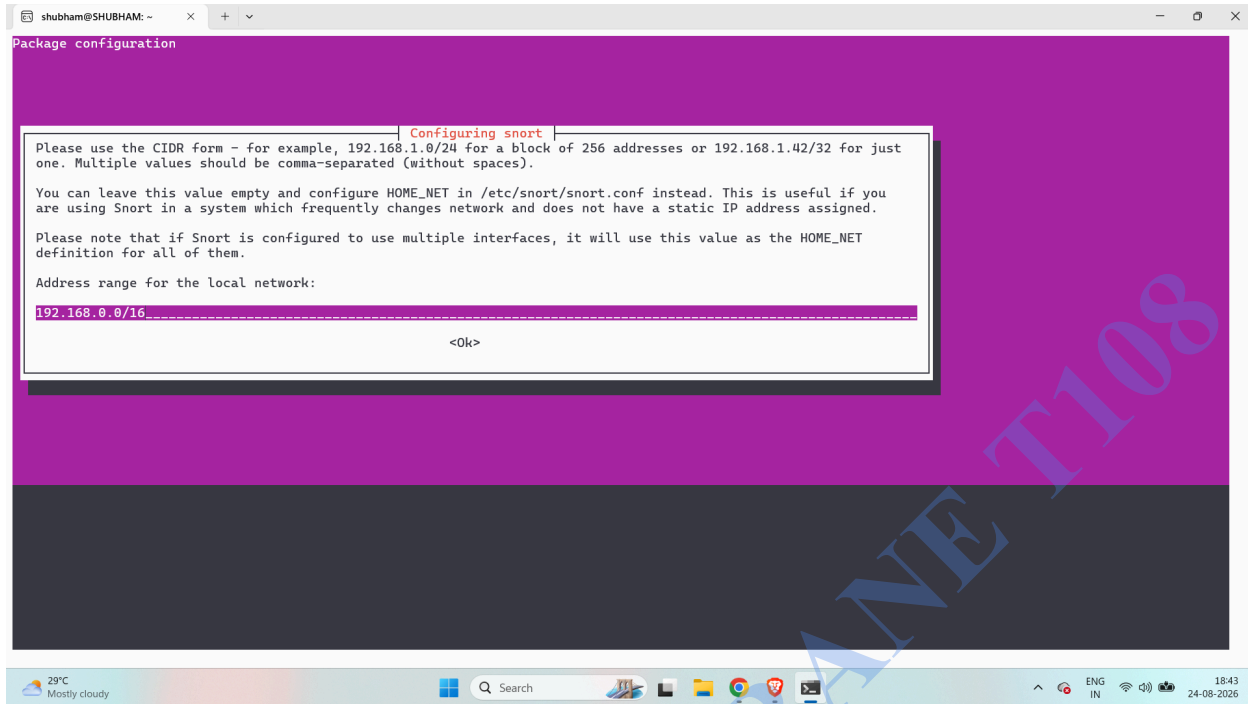
192.168.1.0/24

Enter the network address according to your local network.

```
shubham@SHUBHAM:~$ sudo apt install snort -y
[sudo] password for shubham:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libdrm-nouveau2 libdrm-radeon1 libgl1-amber-dri libglapi-mesa libllvm17t64 libwayland-server0 libxcb-dri2-0
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  ibverbs-providers libauthen-sasl-perl libclone-perl libdaq2t64 libdata-dump-perl libdumbnet1 libencode-locale-perl
  libfile-listing-perl libfont-afm-perl libhtml-form-perl libhtml-format-perl libhtml-parser-perl libhtml-tagset-perl
  libhtml-tree-perl libhttp-cookies-perl libhttp-daemon-perl libhttp-date-perl libhttp-message-perl
  libhttp-negotiate-perl libibverbs1 libio-html-perl libio-socket-ssl-perl liblua5.1-2 liblua5.1-common
  liblwp-mediatypes-perl liblwp-protocol-https-perl libmailtools-perl libnet-http-perl libnet-smtp-ssl-perl
  libnet-ssleay-perl libnetfilter-queue1 libnl-3-200 libnl-route-3-200 libpcap0.8t64 libpcrc3 libtimedate-perl
  libtiny-perl liburi-perl libwww-perl libwww-robotrules-perl net-tools oinkmaster perl-openssl-defaults
  snort-common snort-common-libraries snort-rules-default
Suggested packages:
  libdigest-hmac-perl libgsapi-perl libio-compress-brotli-perl libcrypt-ssleay-perl libsub-name-perl
  libbusiness-isbn-perl librexip-ipv6-perl libauthen-ntlm-perl debhelper snort-doc
The following NEW packages will be installed:
  ibverbs-providers libauthen-sasl-perl libclone-perl libdaq2t64 libdata-dump-perl libdumbnet1 libencode-locale-perl
  libfile-listing-perl libfont-afm-perl libhtml-form-perl libhtml-format-perl libhtml-parser-perl libhtml-tagset-perl
  libhtml-tree-perl libhttp-cookies-perl libhttp-daemon-perl libhttp-date-perl libhttp-message-perl
  libhttp-negotiate-perl libibverbs1 libio-html-perl libio-socket-ssl-perl liblua5.1-2 liblua5.1-common
  liblwp-mediatypes-perl liblwp-protocol-https-perl libmailtools-perl libnet-http-perl libnet-smtp-ssl-perl
  libnet-ssleay-perl libnetfilter-queue1 libnl-3-200 libnl-route-3-200 libpcap0.8t64 libpcrc3 libtimedate-perl
  libtiny-perl liburi-perl libwww-perl libwww-robotrules-perl net-tools oinkmaster perl-openssl-defaults snort
  snort-common snort-common-libraries snort-rules-default
0 upgraded, 47 newly installed, 0 to remove and 3 not upgraded.
Need to get 5305 kB of archives.
After this operation, 19.9 MB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 liblua5.1-2-common all 2.1.0+git20231223.c525bcb+dfsg-1ubuntu0.1 [49.5 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 liblua5.1-2 amd64 2.1.0+git20231223.c525bcb+dfsg-1ubuntu0.1 [276 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble/universe amd64 libpcrc3 amd64 2.8.39-15build1 [248 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble/universe amd64 snort-common-libraries amd64 2.9.20-0+deb11ubuntu1 [899 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble/universe amd64 snort-rules-default all 2.9.20-0+deb11ubuntu1 [144 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble/universe amd64 snort-common all 2.9.20-0+deb11ubuntu1 [47.7 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 net-tools amd64 2.10-0.1ubuntu4.4 [204 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble/universe amd64 libdumbnet1 amd64 1.17.0-1ubuntu2 [30.7 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble/universe amd64 libnetfilter-queue1 amd64 1.0.5-4build1 [15.1 kB]
```

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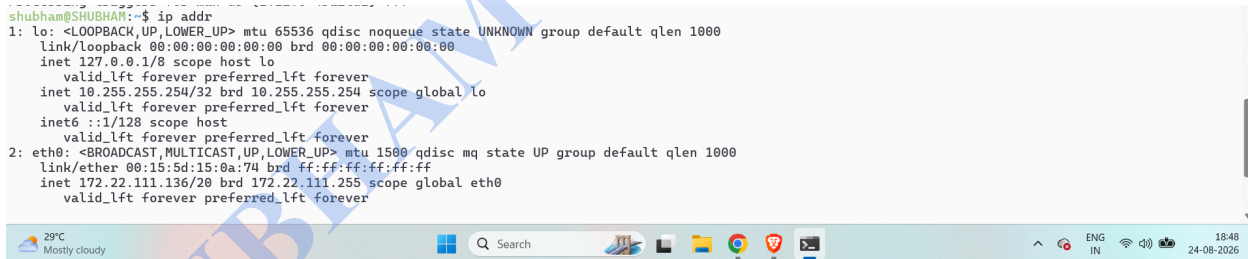


Step 3: Check the Network Interface

Use:

ip addr

Identify the active network interface. Examples: eth0, ens33, enp0s3, wlan0. For this practical, assume the interface is enp0s3.

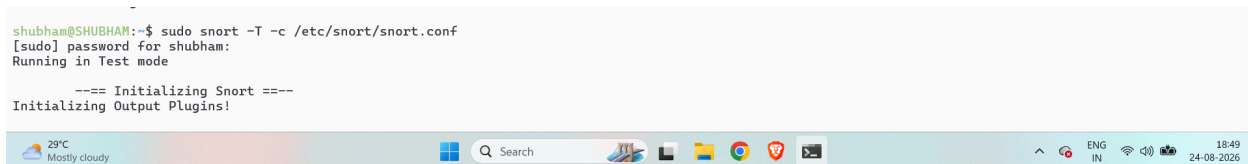


Step 4: Test Snort Configuration

Run:

sudo snort -T -c /etc/snort/snort.conf

The -T option tests the configuration file. If the configuration is correct, Snort should display a successful validation message.



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```
shubham@SHUBHAM: ~  
1 byte states : 1.02  
2 byte states : 13.96  
4 byte states : 0.00  
-----  
[ Number of patterns truncated to 20 bytes: 1038 ]  
MaxRss at the end of detection rules:105168  
--- Initialization Complete ---  
--> Snort! <--  
Version 2.9.20 GRE (Build 82)  
By Martin Roesch & The Snort Team: http://www.snort.org/contact#team  
Copyright (C) 2014-2022 Cisco and/or its affiliates. All rights reserved.  
Copyright (C) 1998-2013 Sourcefire, Inc., et al.  
Using libpcap version 1.10.4 (with TPACKET_V3)  
Using PCRE version: 8.39 2016-06-14  
Using ZLIB version: 1.3  
  
Rules Engine: SF_SNORT_DETECTION_ENGINE Version 3.2 <Build 1>  
Preprocessor Object: SF_REPUTATION Version 1.1 <Build 1>  
Preprocessor Object: appid Version 1.1 <Build 5>  
Preprocessor Object: SF_GTP Version 1.1 <Build 1>  
Preprocessor Object: SF_SSH Version 1.1 <Build 3>  
Preprocessor Object: SF_POP Version 1.0 <Build 1>  
Preprocessor Object: SF_SDF Version 1.1 <Build 1>  
Preprocessor Object: SF_SMTP Version 1.1 <Build 9>  
Preprocessor Object: SF_SIP Version 1.1 <Build 1>  
Preprocessor Object: SF_FTPTELNET Version 1.2 <Build 13>  
Preprocessor Object: SF_SSLPP Version 1.1 <Build 4>  
Preprocessor Object: SF_DNS Version 1.1 <Build 4>  
Preprocessor Object: SF_DNP3 Version 1.1 <Build 1>  
Preprocessor Object: SF_MODBUS Version 1.1 <Build 1>  
Preprocessor Object: SF_S7COMMPLUS Version 1.0 <Build 1>  
Preprocessor Object: SF_IMAP Version 1.0 <Build 1>  
Preprocessor Object: SF_DCERPC2 Version 1.0 <Build 3>  
  
Total snort Fixed Memory Cost - MaxRss:105168  
Snort successfully validated the configuration!  
Snort exiting  
shubham@SHUBHAM:~$ |
```

Step 5: Start Snort in Network Monitoring Mode

Run:

```
sudo snort -i enp0s3 -A console -c /etc/snort/snort.conf
```

Replace enp0s3 with your actual network interface.

- -i specifies the network interface.
- -A console displays alerts on the terminal.
- -c specifies the Snort configuration file.

```
shubham@SHUBHAM: ~  
1 byte states : 1.02  
2 byte states : 13.96  
4 byte states : 0.00  
-----  
[ Number of patterns truncated to 20 bytes: 1038 ]  
pcap DAQ configured to passive.  
Acquiring network traffic from "eth0".  
Reload thread starting...  
Reload thread started, thread 0x7564fa3fe6c8 (1412)  
Decoding Ethernet  
--- Initialization Complete ---  
--> Snort! <--  
Version 2.9.20 GRE (Build 82)  
By Martin Roesch & The Snort Team: http://www.snort.org/contact#team  
Copyright (C) 2014-2022 Cisco and/or its affiliates. All rights reserved.  
Copyright (C) 1998-2013 Sourcefire, Inc., et al.  
Using libpcap version 1.10.4 (with TPACKET_V3)  
Using PCRE version: 8.39 2016-06-14  
Using ZLIB version: 1.3  
  
Rules Engine: SF_SNORT_DETECTION_ENGINE Version 3.2 <Build 1>  
Preprocessor Object: SF_REPUTATION Version 1.1 <Build 1>  
Preprocessor Object: appid Version 1.1 <Build 5>  
Preprocessor Object: SF_GTP Version 1.1 <Build 1>  
Preprocessor Object: SF_SSH Version 1.1 <Build 3>  
Preprocessor Object: SF_POP Version 1.0 <Build 1>  
Preprocessor Object: SF_SDF Version 1.1 <Build 1>  
Preprocessor Object: SF_SMTP Version 1.1 <Build 9>  
Preprocessor Object: SF_SIP Version 1.1 <Build 1>  
Preprocessor Object: SF_FTPTELNET Version 1.2 <Build 13>  
Preprocessor Object: SF_SSLPP Version 1.1 <Build 4>  
Preprocessor Object: SF_DNS Version 1.1 <Build 4>  
Preprocessor Object: SF_DNP3 Version 1.1 <Build 1>  
Preprocessor Object: SF_MODBUS Version 1.1 <Build 1>  
Preprocessor Object: SF_S7COMMPLUS Version 1.0 <Build 1>  
Preprocessor Object: SF_IMAP Version 1.0 <Build 1>  
Preprocessor Object: SF_DCERPC2 Version 1.0 <Build 3>  
  
Commencing packet processing (pid=1411)
```

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Step 6: Create a Custom IDS Rule

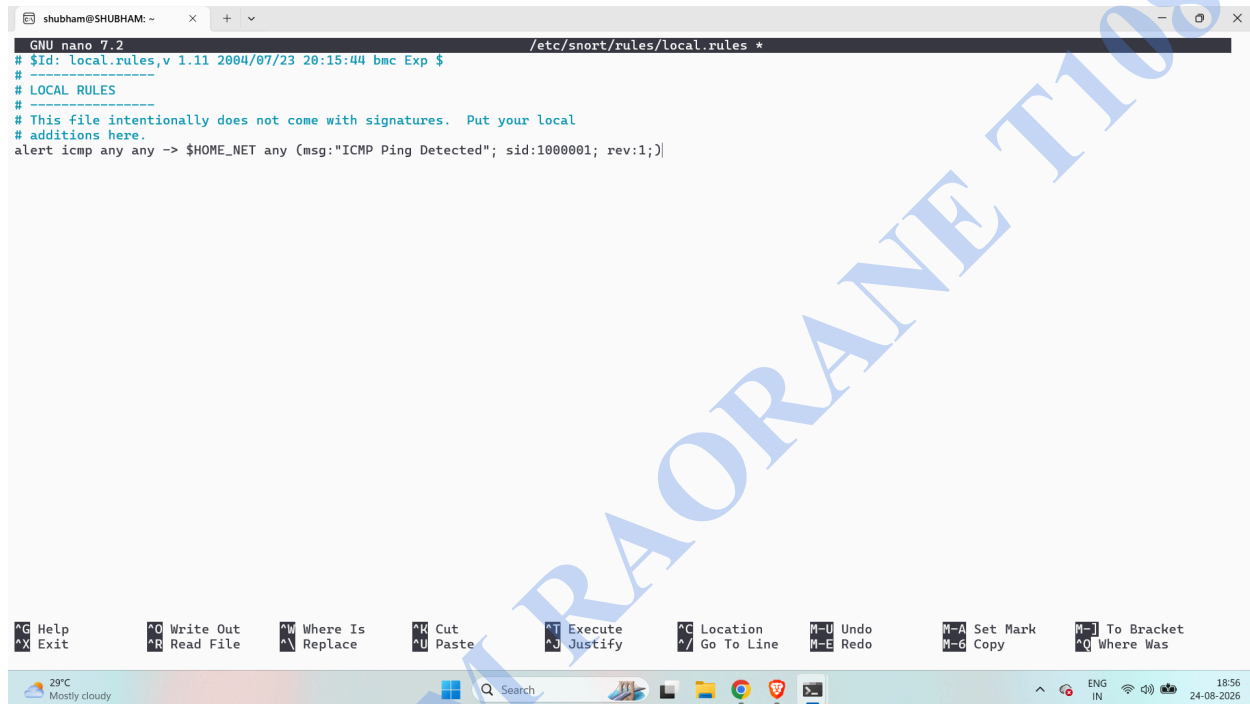
Open the local rules file:

```
sudo nano /etc/snort/rules/local.rules
```

Add:

```
alert icmp any any -> $HOME_NET any (msg:"ICMP Ping Detected";  
sid:1000001; rev:1;)
```

This rule generates an alert whenever ICMP ping traffic is detected.



```
GNU nano 7.2 /etc/snort/rules/local.rules *  
# $Id: local.rules,v 1.11 2004/07/23 20:15:44 bmc Exp $  
#  
# LOCAL RULES  
#  
# This file intentionally does not come with signatures. Put your local  
# additions here.  
alert icmp any any -> $HOME_NET any (msg:"ICMP Ping Detected"; sid:1000001; rev:1;)
```

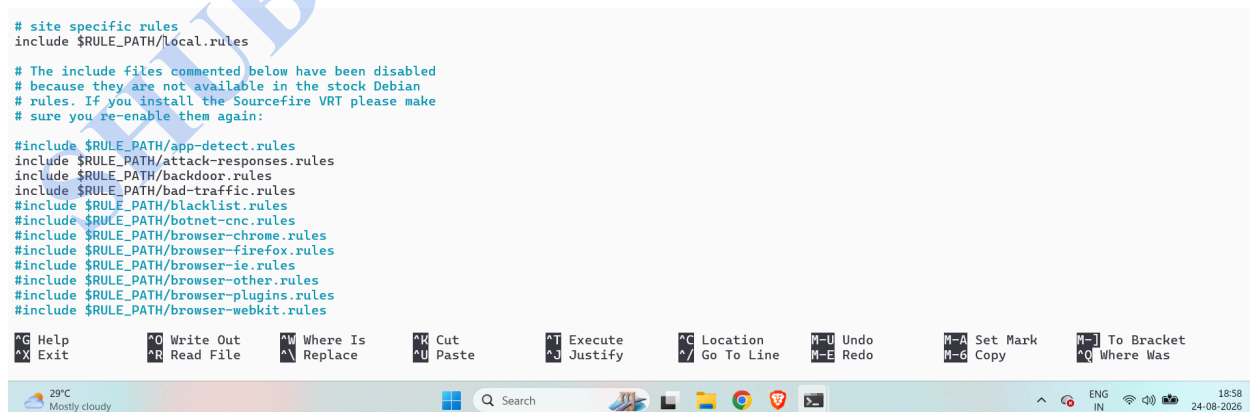
Step 7: Ensure the Local Rule File Is Included

Open:

```
sudo nano /etc/snort/snort.conf
```

Check that this line is present:

```
include $RULE_PATH/local.rules
```



```
# site specific rules  
include $RULE_PATH/local.rules  
  
# The include files commented below have been disabled  
# because they are not available in the stock Debian  
# rules. If you install the Sourcefire VRT please make  
# sure you re-enable them again:  
  
#include $RULE_PATH/app-detect.rules  
include $RULE_PATH/attack-responses.rules  
include $RULE_PATH/backdoor.rules  
include $RULE_PATH/bad-traffic.rules  
#include $RULE_PATH/blacklist.rules  
#include $RULE_PATH/botnet-cnc.rules  
#include $RULE_PATH/browser-chrome.rules  
#include $RULE_PATH/browser-firefox.rules  
#include $RULE_PATH/browser-ie.rules  
#include $RULE_PATH/browser-other.rules  
#include $RULE_PATH/browser-plugins.rules  
#include $RULE_PATH/browser-webkit.rules
```

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Step 8: Test the Configuration Again

Run:

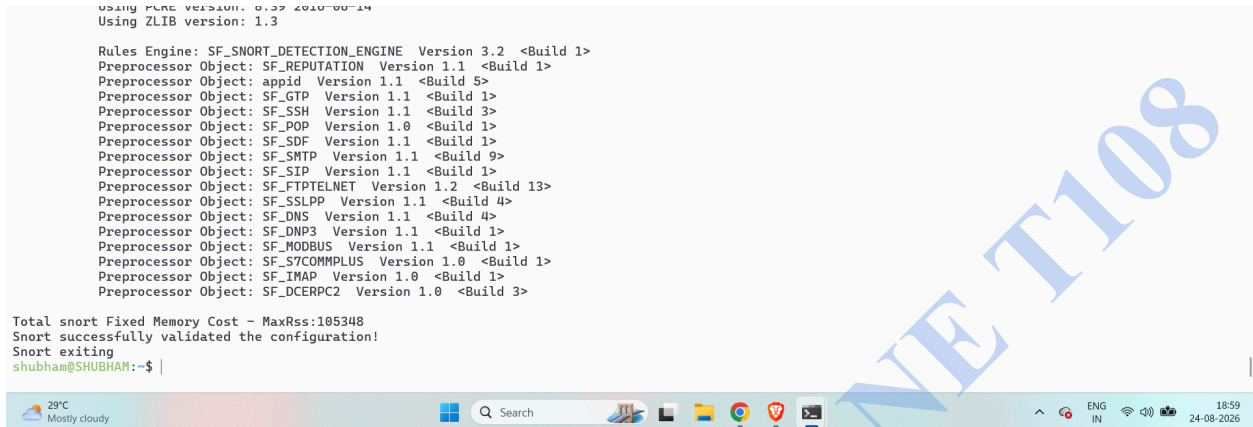
```
sudo snort -T -c /etc/snort/snort.conf
```

Correct any errors before continuing.

```
USING PCRE VERSION: 8.37 2019-08-14
Using ZLIB version: 1.3

Rules Engine: SF_SNORT_DETECTION_ENGINE Version 3.2 <Build 1>
Preprocessor Object: SF_REPUTATION Version 1.1 <Build 1>
Preprocessor Object: appid Version 1.1 <Build 5>
Preprocessor Object: SF_GTP Version 1.1 <Build 1>
Preprocessor Object: SF_SSH Version 1.1 <Build 3>
Preprocessor Object: SF_POP Version 1.0 <Build 1>
Preprocessor Object: SF_SDF Version 1.1 <Build 1>
Preprocessor Object: SF_SMTP Version 1.1 <Build 9>
Preprocessor Object: SF_SIP Version 1.1 <Build 1>
Preprocessor Object: SF_PTPTNET Version 1.2 <Build 13>
Preprocessor Object: SF_SSLLP Version 1.1 <Build 4>
Preprocessor Object: SF_DNS Version 1.1 <Build 4>
Preprocessor Object: SF_DNP3 Version 1.1 <Build 1>
Preprocessor Object: SF_MODBUS Version 1.1 <Build 1>
Preprocessor Object: SF_S7COMMPPLUS Version 1.0 <Build 1>
Preprocessor Object: SF_IMAP Version 1.0 <Build 1>
Preprocessor Object: SF_DCERPC2 Version 1.0 <Build 3>

Total snort Fixed Memory Cost - MaxRss:105348
Snort successfully validated the configuration!
Snort exiting
shubham@SHUBHAM:~$ |
```



Step 9: Start IDS Monitoring

Run:

```
sudo snort -i enp0s3 -A console -c /etc/snort/snort.conf
```

The IDS is now actively monitoring network traffic.

Step 10: Generate Test Traffic

From another authorized computer on the same network:

ping <IDS-IP-ADDRESS>;

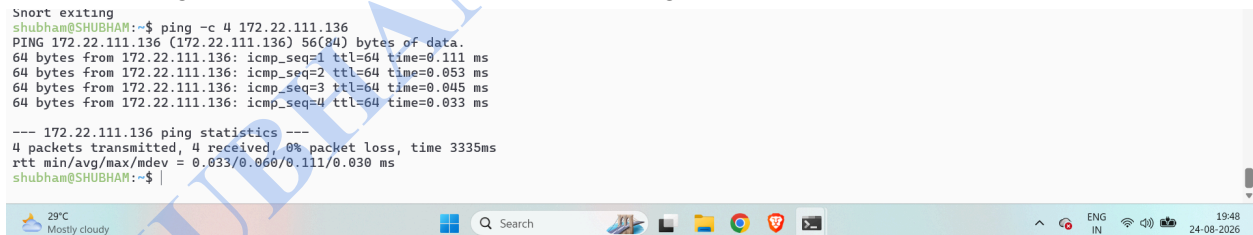
Example:

ping 192.168.1.10

Snort should generate an alert similar to: ICMP Ping Detected.

```
Snort exiting
shubham@SHUBHAM:~$ ping -c 4 172.22.111.136
PING 172.22.111.136 (172.22.111.136) 56(84) bytes of data.
64 bytes from 172.22.111.136: icmp_seq=1 ttl=64 time=0.111 ms
64 bytes from 172.22.111.136: icmp_seq=2 ttl=64 time=0.053 ms
64 bytes from 172.22.111.136: icmp_seq=3 ttl=64 time=0.045 ms
64 bytes from 172.22.111.136: icmp_seq=4 ttl=64 time=0.033 ms

--- 172.22.111.136 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 333ms
rtt min/avg/max/mdev = 0.033/0.060/0.111/0.030 ms
shubham@SHUBHAM:~$ |
```



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Step 11: Test Port Scanning Detection

Install Nmap:

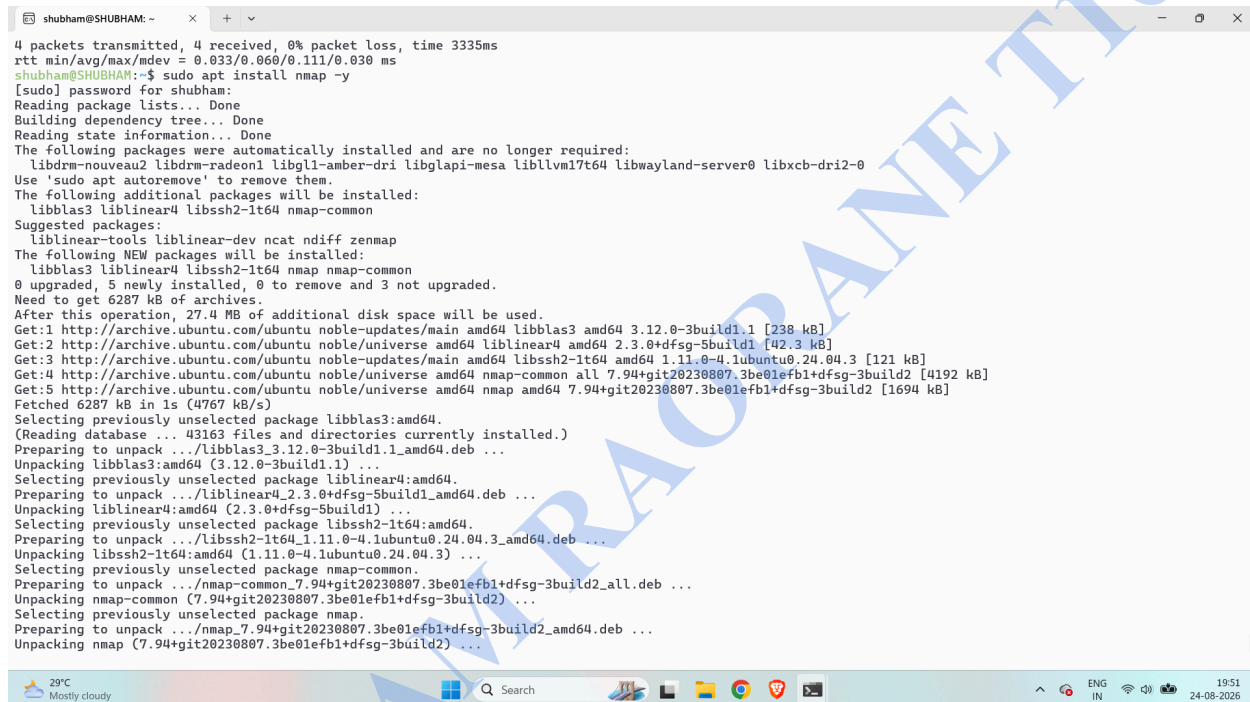
sudo apt install nmap -y

Add the following rule to /etc/snort/rules/local.rules:

```
alert tcp any any -> $HOME_NET any (flags:S; msg:"Possible TCP SYN Scan  
Detected";  
sid:1000002; rev:1;)
```

Restart Snort. From an authorized lab machine:

nmap -sS <TARGET-IP-ADDRESS>;



```
shubham@SHUBHAM: ~  
4 packets transmitted, 4 received, 0% packet loss, time 3335ms  
rtt min/avg/max/mdev = 0.033/0.060/0.111/0.030 ms  
shubham@SHUBHAM:~$ sudo apt install nmap -y  
[sudo] password for shubham:  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
The following packages were automatically installed and are no longer required:  
  libdrm-nouveau2 libdrm-radeon1 libgl1-amber-dri libglapi-mesa liblvm17t64 libwayland-server0 libxcb-dri2-0  
Use 'sudo apt autoremove' to remove them.  
The following additional packages will be installed:  
  libblas3 liblinear4 libssh2-1t64 nmap nmap-common  
Suggested packages:  
  liblinear-tools liblinear-dev ncat ndiff zenmap  
The following NEW packages will be installed:  
  libblas3 liblinear4 libssh2-1t64 nmap nmap-common  
0 upgraded, 5 newly installed, 0 to remove and 3 not upgraded.  
Need to get 6287 kB of archives.  
After this operation, 27.4 MB of additional disk space will be used.  
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libblas3 amd64 3.12.0-3build1.1 [238 kB]  
Get:2 http://archive.ubuntu.com/ubuntu noble/universe amd64 liblinear4 amd64 2.3.0+dfsg-5build1 [42.3 kB]  
Get:3 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libssh2-1t64 amd64 1.11.0-4.1ubuntu0.24.04.3 [121 kB]  
Get:4 http://archive.ubuntu.com/ubuntu noble/universe amd64 nmap-common all 7.94+git20230807.3be01efb1+dfsg-3build2 [4192 kB]  
Get:5 http://archive.ubuntu.com/ubuntu noble/universe amd64 nmap amd64 7.94+git20230807.3be01efb1+dfsg-3build2 [1694 kB]  
Fetched 6287 kB in 1s (4767 kB/s)  
Selecting previously unselected package libblas3:amd64.  
(Reading database ... 43163 files and directories currently installed.)  
Preparing to unpack .../libblas3_3.12.0-3build1.1_amd64.deb ...  
Unpacking libblas3:amd64 (3.12.0-3build1.1) ...  
Selecting previously unselected package liblinear4:amd64.  
Preparing to unpack .../liblinear4_2.3.0+dfsg-5build1_amd64.deb ...  
Unpacking liblinear4:amd64 (2.3.0+dfsg-5build1) ...  
Selecting previously unselected package libssh2-1t64:amd64.  
Preparing to unpack .../libssh2-1t64_1.11.0-4.1ubuntu0.24.04.3_amd64.deb ...  
Unpacking libssh2-1t64:amd64 (1.11.0-4.1ubuntu0.24.04.3) ...  
Selecting previously unselected package nmap-common.  
Preparing to unpack .../nmap-common_7.94+git20230807.3be01efb1+dfsg-3build2_all.deb ...  
Unpacking nmap-common (7.94+git20230807.3be01efb1+dfsg-3build2) ...  
Selecting previously unselected package nmap.  
Preparing to unpack .../nmap_7.94+git20230807.3be01efb1+dfsg-3build2_amd64.deb ...  
Unpacking nmap (7.94+git20230807.3be01efb1+dfsg-3build2) ...
```

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The screenshot shows a Windows terminal window with two panels. The top panel displays the contents of the file `/etc/snort/rules/local.rules` using the `nano` editor. The file contains local rules for detecting ICMP ping and TCP SYN scans. The bottom panel shows the execution of `sudo nmap -sS 172.22.111.136`, which returns a successful scan report for the target IP address, identifying open ports 80 (http) and 443 (https).

```
GNU nano 7.2 /etc/snort/rules/local.rules
# $Id: local.rules,v 1.11 2004/07/23 20:15:44 bmc Exp $
#
# LOCAL RULES
#
# This file intentionally does not come with signatures. Put your local
# additions here.
alert icmp any any -> $HOME_NET any (msg:"ICMP Ping Detected"; sid:1000001; rev:1;)
alert tcp any any -> $HOME_NET any (flags:S; msg:"Possible TCP SYN Scan Detected"; sid:1000002; rev:1;)
```

```
shubham@SHUBHAM:~$ sudo nmap -sS 172.22.111.136
[sudo] password for shubham:
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-08-24 14:26 UTC
Nmap scan report for 172.22.111.136
Host is up (0.0000010s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE
80/tcp    open  http
443/tcp   open  https

Nmap done: 1 IP address (1 host up) scanned in 0.10 seconds
shubham@SHUBHAM:~$
```

Step 12: View IDS Alerts

Alerts may be displayed directly when using -A console. Logs can also be checked in:

`/var/log/snort/`

Commands:

`sudo ls /var/log/snort/`

`sudo cat /var/log/snort/alert`

The screenshot shows a Windows terminal window with the command `sudo ls -l /var/log/snort/` executed, listing three files: `snort.alert.fast`, `snort.log.1787578183`, and `snort.log.1787581551`. The subsequent command `sudo cat /var/log/snort/alert` results in an error message: "cat: /var/log/snort/alert: No such file or directory".

```
shubham@SHUBHAM:~$ sudo ls -l /var/log/snort/
total 128
-rw-r--r-- 1 root adm 34924 Aug 24 14:17 snort.alert.fast
-rw-r--r-- 1 root adm 88536 Aug 24 14:17 snort.log.1787578183
-rw-r--r-- 1 root adm    0 Aug 24 14:25 snort.log.1787581551
shubham@SHUBHAM:~$ sudo cat /var/log/snort/alert
cat: /var/log/snort/alert: No such file or directory
shubham@SHUBHAM:~$
```

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Step 13: Monitor Traffic Using Wireshark

Install Wireshark:

```
sudo apt install wireshark -y
```

Launch:

wireshark

Select the active interface and start capturing packets. Useful filters include: icmp and tcp.



```
shubham@SHUBHAM: ~$ sudo apt install wireshark -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libdrm-nouveau2 libdrm-radeon1 libgl1-amber-dri libglapi-mesa libllvm17t64 libxcb-dri2-0
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  gststreamer1.0-glib gststreamer1.0-plugins-base i965-via-driver intel-media-via-driver libaacs0 libaom3 libasynccs0 libavcodec60 libavformat60 libavutil58
  libb2-1 libbcb729-0 libbdplus0 libbluray2 libbcs2 libcdparanoia0 libchromaprint1 libcsound1 libcodecs2-1.2 libdavid7 libdouble-conversion3 libdev2
  libflac12t64 libgme0 libgomp1 libgraphene-1.0-0 libgsm1 libgststreamer-glib1.0-0 libgststreamer-plugins-base1.0-0 libgudev-1.0-0 libhwy1t64 libice6
  libidn2 libinput-bin libinput0 libjx10.7 liblua5.2-0 libmaxminddb0 libmbc2crypto7t64 libmd4c0 libminizip1t64 libmp3lame0 libmpg123-0t64
  libmtdev1t64 libnghttp3-3 libnl-genl-3-200 libnorm1t64 libnuma1 libogg0 libopencl-amrnb0 libopencl0 libopenjp2-7 libopenmpt0t64 libopus0
  liborc-0.4-0t64 libpcre2-16-0 libpgm-5.3-0t64 libproxy1v5 libpulse0 libqt6core5compat6 libqt6core6t64 libqt6dbus6t64 libqt6gui6t64 libqt6multimedia6
  libqt6network6t64 libqt6opengl6t64 libqt6printsupport6t64 libqt6qml6 libqt6qmlmodels6 libqt6quick6 libqt6svg6 libqt6waylandclient6
  libqt6waylandcompositor6 libqt6waylandeglclienthwintegration6 libqt6waylandeglcompositorhwintegration6 libqt6widgets6t64 libqt6wllshellintegration6
  librabbitmq4 libravie0 librist4 libsbcl1 libshine3 libsm6 libsmi2t64 libsnappy1v5 libsndfile1 libsoxr0 libspandsp2t64 libspeex1 libspeexdsp1
  libstr1.5-gnutls libssh-gcrypt-4 libsvtav1enc1d1 libswresample4 libswscale7 libtheora0 libts0t64 libtwolame0 libudfread0 libva-drm2 libva-x11-2 libva2
  libvdpau1 libvisual-0.4-0 libvorbis0a libvorbisenc2 libvorbisfile3 libvpl2 libvpx9 libwacom-common libwacom9 libwebp0 libwebpmux3 libwireshark-data
  libwireshark17t64 libwiretap14t64 libwsutil15t64 libx264-164 libx265-199 libxcb-iccmm4 libxcb-image0 libxcb-keysyms1 libxcb-render-util0 libxcb-shape0
  libxcb-util1 libxcb-xkb1 libxkbcommon-x11-0 libxvidcore4 libxvms5 libzvi-common libzvi0t64 mesa-via-drivers mesa-vdpau-drivers ocl-icd-libopencl1
  qt6-gtk-platformtheme qt6-qtbase-plugins qt6-translations-l10n qt6-wayland va-driver-all vdpau-driver-all wireshark-common
Suggested packages:
  gvfs i965-via-driver-shaders libcdudal libnvcuvid1 libnvidia-encode1 libbluray-bdj libvisual-0.4-plugins mmd-bins opus-tools pulseaudio
  qt6-qmltooling-plugins snmp-mibs-downloader speex libwacom-bin geopipupdate geopip-database geopip-database-extra libjs-leaflet libjs-leaflet.markercluster
  wireshark-doc opencl-icd libvdpau-via-gli
The following NEW packages will be installed:
  gststreamer1.0-glib gststreamer1.0-plugins-base i965-via-driver intel-media-via-driver libaacs0 libaom3 libasynccs0 libavcodec60 libavformat60 libavutil58
  libb2-1 libbcb729-0 libbdplus0 libbluray2 libbcs2 libcdparanoia0 libchromaprint1 libcsound1 libcodecs2-1.2 libdavid7 libdouble-conversion3 libdev2
  libflac12t64 libgme0 libgomp1 libgraphene-1.0-0 libgsm1 libgststreamer-glib1.0-0 libgststreamer-plugins-base1.0-0 libgudev-1.0-0 libhwy1t64 libice6
  libidn2 libinput-bin libinput0 libjx10.7 liblua5.2-0 libmaxminddb0 libmbc2crypto7t64 libmd4c0 libminizip1t64 libmp3lame0 libmpg123-0t64
  libmtdev1t64 libnghttp3-3 libnl-genl-3-200 libnorm1t64 libnuma1 libogg0 libopencl-amrnb0 libopencl0 libopenjp2-7 libopenmpt0t64 libopus0
  liborc-0.4-0t64 libpcre2-16-0 libpgm-5.3-0t64 libproxy1v5 libpulse0 libqt6core5compat6 libqt6core6t64 libqt6dbus6t64 libqt6gui6t64 libqt6multimedia6
  libqt6network6t64 libqt6opengl6t64 libqt6printsupport6t64 libqt6qml6 libqt6qmlmodels6 libqt6quick6 libqt6svg6 libqt6waylandclient6
  libqt6waylandcompositor6 libqt6waylandeglclienthwintegration6 libqt6waylandeglcompositorhwintegration6 libqt6widgets6t64 libqt6wllshellintegration6
  librabbitmq4 libravie0 librist4 libsbcl1 libshine3 libsm6 libsmi2t64 libsnappy1v5 libsndfile1 libsoxr0 libspandsp2t64 libspeex1 libspeexdsp1
  libstr1.5-gnutls libssh-gcrypt-4 libsvtav1enc1d1 libswresample4 libswscale7 libtheora0 libts0t64 libtwolame0 libudfread0 libva-drm2 libva-x11-2 libva2
  libvdpau1 libvisual-0.4-0 libvorbis0a libvorbisenc2 libvorbisfile3 libvpl2 libvpx9 libwacom-common libwacom9 libwebp0 libwebpmux3 libwireshark-data
  libwireshark17t64 libwiretap14t64 libwsutil15t64 libx264-164 libx265-199 libxcb-iccmm4 libxcb-image0 libxcb-keysyms1 libxcb-render-util0 libxcb-shape0
  libxcb-util1 libxcb-xkb1 libxkbcommon-x11-0 libxvidcore4 libxvms5 libzvi-common libzvi0t64 mesa-via-drivers mesa-vdpau-drivers ocl-icd-libopencl1
  qt6-gtk-platformtheme qt6-qtbase-plugins qt6-translations-l10n qt6-wayland va-driver-all vdpau-driver-all wireshark-common
```

Package configuration

Configuring wireshark-common

Dumpcap can be installed in a way that allows members of the "wireshark" system group to capture packets. This is recommended over the alternative of running Wireshark/Tshark directly as root, because less of the code will run with elevated privileges.

For more detailed information please see /usr/share/doc/wireshark-common/README.Debian.gz once the package is installed.

Enabling this feature may be a security risk, so it is disabled by default. If in doubt, it is suggested to leave it disabled.

Should non-superusers be able to capture packets?

<Yes> <No>

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SUBJECT : CYBER SECURITY

```
shubham@SHUBHAM:~$ sudo tshark -i eth0 -f "icmp"
```

```
Running as user "root" and group "root". This could be dangerous.
```

```
Capturing on 'eth0'
```

```
1 0.000000000 172.22.111.136 → 172.22.96.1 ICMP 98 Echo (ping) request id=0x0dff, seq=1/256, ttl=64
2 0.000472760 172.22.96.1 → 172.22.111.136 ICMP 98 Echo (ping) reply id=0x0dff, seq=1/256, ttl=128 (request in 1)
3 3.934183512 172.22.111.136 → 172.22.96.1 ICMP 98 Echo (ping) request id=0x0dff, seq=2/512, ttl=64
4 3.934811309 172.22.96.1 → 172.22.111.136 ICMP 98 Echo (ping) reply id=0x0dff, seq=2/512, ttl=128 (request in 3)
5 4.958104172 172.22.111.136 → 172.22.96.1 ICMP 98 Echo (ping) request id=0x0dff, seq=3/768, ttl=64
6 4.958450333 172.22.96.1 → 172.22.111.136 ICMP 98 Echo (ping) reply id=0x0dff, seq=3/768, ttl=128 (request in 5)
7 5.982318565 172.22.111.136 → 172.22.96.1 ICMP 98 Echo (ping) request id=0x0dff, seq=4/1024, ttl=64
8 5.983158111 172.22.96.1 → 172.22.111.136 ICMP 98 Echo (ping) reply id=0x0dff, seq=4/1024, ttl=128 (request in 7)
```

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Preprocessor Object: St_UCLAMP_C2 Version 1.0 <Build 3>
Commencing packet processing (pid=3606)

```
08/24-15:07:03.672177 [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 172.22.111.136 -> 172.22.96.1
08/24-15:07:03.672668 [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 172.22.96.1 -> 172.22.111.136
08/24-15:07:04.698241 [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 172.22.111.136 -> 172.22.96.1
08/24-15:07:04.698852 [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 172.22.96.1 -> 172.22.111.136
08/24-15:07:05.722167 [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 172.22.111.136 -> 172.22.96.1
08/24-15:07:05.722717 [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 172.22.96.1 -> 172.22.111.136
08/24-15:07:06.746200 [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 172.22.111.136 -> 172.22.96.1
08/24-15:07:06.747113 [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 172.22.96.1 -> 172.22.111.136
08/24-15:08:38.918863 [**] [1:1917:6] SCAN UPnP service discover attempt [**] [Classification: Detection of a Network Scan] [Priority: 3] {UDP} 172.22.96.1:58542 -> 239.255.255.250:1900
08/24-15:08:38.955956 [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {IPv6-ICMP} fe80::286a:149b:fbd3:cf97 -> ff02::16
```

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20:35 24-08-2026